RESEARCH ARTICLE

Functional Outcomes and Safety of Endovascular Thrombectomy in Pediatric Acute Ischemic Stroke With Cardiac Abnormalities

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Neurology® 2026;107:e218270. doi:10.1212/WNL.0000000000218270

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Abstract

Background and Objectives
Cardiac abnormalities are among the most common and complex risk factors of pediatric acute ischemic stroke (AIS). Although endovascular treatment (EVT) is increasingly used in children, uncertainty persists regarding its safety and effectiveness in patients with underlying cardiac disease, leading to therapeutic hesitation in this high-risk population. The aim of this study was to determine whether the association between EVT and functional outcomes after pediatric AIS differs according to cardiac abnormality status.

Methods
This study was a retrospective secondary analysis of a prospective, multinational cohort registry (Save ChildS Pro). The registry enrolled patients between 2018 and 2023, with 90-day follow-up. Participants included children aged 28 days to 18 years with confirmed AIS due to large vessel occlusion and available 90-day Pediatric Stroke Outcome Measure (PSOM) scores. Cardiac abnormalities were defined as congenital or acquired cardiac disease documented by echocardiography or medical records. The primary outcome was poor functional outcome at 90 days, defined as PSOM score >0.5. Secondary outcomes included ordinal 90-day modified Rankin Scale (mRS) scores and longitudinal neurologic recovery assessed using the Pediatric NIH Stroke Scale (pedNIHSS). Treatment-subgroup interaction was assessed using penalized logistic regression.

Results
Of 208 eligible patients, 178 were included in the final analysis after excluding those with missing outcome data [43% female; mean age: 11.5 years]. Among 178 children, 56 (31.5%) had cardiac abnormalities. EVT was more frequently performed in children with cardiac abnormalities than in those without (p = 0.003). Cardiac abnormality status was not independently associated with outcome (odds ratio, 0.93; 95% CI, 0.36–2.52), and no significant interaction between EVT and cardiac abnormality was observed. Ordinal mRS shift analysis and longitudinal pedNIHSS modeling supported similar recovery trajectories. Adverse event rates did not differ by EVT status in either subgroup.

Discussion
In this multinational pediatric AIS cohort, EVT was associated with improved functional outcomes and accelerated neurologic recovery, with no evidence that cardiac abnormality status modifies association of treatment with outcome or safety. These findings support consideration of EVT in children with cardiac abnormalities when clinically indicated.

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Glossary

AIS = acute ischemic stroke; **ASPECTS** = Alberta Stroke Program Early CT Score; **EVT** = endovascular treatment; **mRS** = modified Rankin Scale; **ORs** = odds ratio; **pedNIHSS** = Pediatric NIH Stroke Scale; **SMD** = standardized mean difference; **PSOM** = Pediatric Stroke Outcome Measure.

Classification of Evidence

This study provides class IV evidence that endovascular therapy in pediatric patients with AIS is associated with improved functional outcome irrespective of underlying cardiac abnormality status.

Introduction

Pediatric acute ischemic stroke (AIS) is a rare but highly consequential neurologic emergency, frequently resulting in persistent motor, cognitive, and psychosocial impairment.[1] Compared with adult stroke, pediatric AIS is characterized by greater etiologic heterogeneity, delayed recognition, and limited high-quality evidence to guide acute management.[2] Among the recognized causes, cardiac abnormalities, including congenital heart disease and acquired cardiac conditions, constitute one of the most prevalent and clinically complex risk factors of childhood AIS.[3] Children with cardiac disease often experience embolic strokes, may present at younger ages, and frequently have substantial comorbidities that complicate both acute treatment and recovery.[4]

The role of endovascular treatment (EVT) in pediatric stroke remains an area of active investigation.[5] Although EVT has transformed outcomes in adults with large vessel occlusion, its application in children relies primarily on observational data and expert consensus.[6] It is important to note that children with cardiac abnormalities have historically raised particular concern regarding EVT candidacy. These concerns include altered clot composition, higher embolic risk, vascular access challenges, anticoagulation status, periprocedural hemodynamic instability, and perceived risks of hemorrhage or procedural complications.[7,8]

Given these considerations, it remains unclear whether EVT confers comparable functional benefit in children with cardiac abnormalities relative to those without cardiac disease. From a clinical perspective, even demonstrating no decrement in outcomes among children with cardiac abnormalities would carry substantial significance. However, existing pediatric EVT studies have rarely examined cardiac status as a potential effect modifier, and most have reported aggregate outcomes without stratified or interaction-based analyses.[6,9]

In this study, we investigated the association between EVT and functional outcomes in a pediatric AIS cohort, with a specific focus on whether cardiac abnormality status modifies the association of EVT with outcome. To address challenges inherent to pediatric stroke research, including limited sample size and group imbalance, we applied penalized regression methods alongside longitudinal and matched analyses. By explicitly examining treatment-subgroup interactions, our aim was to clarify the clinical impact of EVT in children with cardiac abnormalities and to inform risk-benefit considerations in this population.

Methods

This was an analysis of the Prospective Multicenter Registry on Feasibility, Safety, and Outcome of Endovascular Recanalization in Childhood Stroke (Save ChildS Pro) study,[10] a multinational cohort registry of pediatric AIS in North America, South America, Europe, Asia, and Australia. The supporting data are available from the corresponding author on reasonable request. This study was conducted based on the Strengthening the Reporting of Observational Studies in Epidemiology guidelines.[11] Approval was obtained from the institutional review boards.

Study Population

Eligible participants were pediatric patients enrolled in the Save ChildS Pro registry who met the following inclusion criteria: (1) had a large vessel occlusion, defined as occlusion of the internal carotid artery, MCA-M1, ACA-A1, basilar artery, vertebral artery, or PCA-P; (2) had a confirmed clinical diagnosis of AIS; and (3) had an available 90-day (±10 days) Pediatric Stroke Outcome Measure (PSOM) score. All included patients received best medical therapy according to local institutional protocols. Based on treatment decisions made by the treating multidisciplinary stroke teams at each center, a subset of patients further underwent EVT. Patients were classified as having a cardiac abnormality if structural or functional cardiac pathology was identified on echocardiography or documented in the medical record. The cardiac abnormality was considered a contributing factor to the stroke by the treatment team.

Data Collection

Local investigators entered data into the registry, including baseline demographic and clinical variables: age, sex, antithrombotic use (at baseline), stroke circulation territory, occlusion site, length of hospital stay, anesthesia type, number of thrombectomy passes (when applicable), and use of IV thrombolysis or anticoagulation. Imaging severity was assessed locally using the Alberta Stroke Program Early CT Score (ASPECTS) where applicable.

Neurologic and functional outcomes were locally evaluated using standardized, validated pediatric stroke scales. Stroke severity was assessed using the Pediatric NIH Stroke Scale (pedNIHSS) at admission, 12–24 hours, and discharge. Functional outcomes were measured using the PSOM and the modified Rankin Scale (mRS) at discharge and 90 (±10) days. Adverse events tracked included need for decompressive hemicraniectomy, need for extraventricular drainage, intracerebral hemorrhage, and recurrent stroke.

The primary outcome in this study was poor 90-day functional outcome, operationalized as a binary end point derived from the PSOM score at 90 days at a threshold of 0.5. Secondary outcomes included (1) 90-day mRS score as an ordinal end point for shift analysis and (2) longitudinal change in neurologic severity where available (pedNIHSS scores at baseline and follow-up time points), used descriptively for trajectory visualization and exploratory modeling.

Statistical Analysis

To quantify the association between EVT and poor 90-day outcome and to test whether this association differed by cardiac abnormality status, we fit a penalized (Firth) logistic regression model with the prespecified interaction term EVT × cardiac abnormality, adjusting for clinically relevant covariates (age, baseline pedNIHSS score, and ASPECTS). Firth penalized maximum likelihood estimation was chosen a priori to reduce small-sample and separation-related bias that can occur when modeling treatment effects and interactions in modest sample sizes and unbalanced treatment groups. Model results were reported as odds ratios (ORs) with 95% CIs. For regression analyses, missing values in continuous predictors were handled using median imputation within the modeling pipeline.

In addition, we performed stratified analyses within the cardiac abnormality and no–cardiac abnormality groups to estimate EVT-associated ORs within each stratum, using similarly adjusted penalized logistic models to support interpretability of treatment effects across subgroups.

To evaluate treatment and cardiac abnormality associations across the full distribution of disability, we used an ordinal logistic regression (proportional odds) model with 90-day mRS scores as the dependent variable. The model included EVT, cardiac abnormality, their interaction, and covariates (age, baseline pedNIHSS score, and ASPECTS). Results were presented as log-odds coefficients (and/or proportional ORs where applicable), with 2-sided *p* values.

Because EVT uptake differed by cardiac abnormality status, we conducted propensity score matching to reduce measured confounding when comparing groups.[10,12] Propensity scores were estimated using logistic regression with cardiac abnormality as the exposure and baseline covariates selected for clinical relevance and availability (age, sex, baseline pedNIHSS score, ASPECTS, and circulation [anterior vs posterior]). We then applied nearest-neighbor matching with a 1:2 ratio and a caliper of 0.2. Covariate balance was assessed using standardized mean differences (SMDs), targeting SMD <0.1 as evidence of acceptable postmatch balance; balance diagnostics and matched sample sizes were reported.

All tests were two-sided. Continuous variables were summarized using medians and interquartile ranges because of nonnormal distributions. Effect estimates were reported with 95% CIs. Analyses were implemented using standard statistical software (R for penalized logistic regression and matching; Python/stats libraries for supporting analyses and visualization), and all analysis decisions (covariate set, interaction testing, and propensity score matching specifications) were defined to align with the clinical question of whether EVT benefit differed by cardiac abnormality status.

Standard Protocol Approvals, Registrations, and Patient Consents

The Save ChildS Pro registry was approved by the institutional review board/ethics committee at each participating center, and this retrospective analysis of deidentified registry data was conducted under those approvals. Written informed consent for participation in the registry was obtained from participants' parents/guardians (and assent from participants when applicable) in accordance with local regulations; where permitted, consent was waived by the local ethics committee for retrospective analysis of deidentified data. No identifiable photographs, videos, or other potentially recognizable information are included in this article; therefore, consent for disclosure was not required.

Data Availability

Deidentified individual participant data underlying the results reported in this article, along with the statistical analysis plan, will be made available on reasonable request after publication. Data will be shared with qualified researchers for noncommercial research purposes, subject to approval by the corresponding author and after completion of a data use agreement. Requests should include a brief proposal describing the intended analyses. Access will be provided through secure transfer, and the data will be available for 1 year after publication.

Results

Study Population and Baseline Characteristics

A total of 178 pediatric patients with AIS were included in the initial cohort, of whom 56 (31.5%) had a documented cardiac abnormality (Figure 1). Patients with cardiac abnormalities were younger (median age 7.5 [interquartile range 3.0–12.0] vs 9.0 [6.0–15.0] years, $p = 0.020$) and were more likely to undergo EVT (76.8% vs 52.5%, $p = 0.003$). They also experienced longer hospital stays (median 15 vs 12 days, $p =$

0.006). Baseline stroke severity, measured by pedNIHSS scores at admission, ASPECTS, and early PSOM scores, did not differ significantly between groups (Table 1 and Figure 2).

To mitigate baseline imbalances and reduce confounding by indication, propensity score matching was performed using nearest-neighbor matching with a 1:2 ratio and a caliper of 0.2. Covariates included age, sex, baseline pedNIHSS score, ASPECTS, and circulation territory. After matching, the cohort comprised 52 patients with cardiac abnormalities and 85 without cardiac abnormalities. All covariates achieved adequate balance, with SMDs <0.1, indicating successful matching (eFigure 1). Baseline clinical severity, imaging characteristics, time to treatment, and demographic variables were comparable between groups in the matched cohort. In the matched cohort, EVT remained significantly more frequent in children with cardiac abnormalities compared with those without (80.8% vs 51.8%, $p < 0.001$). The number of thrombectomy passes remained higher in the cardiac abnormality group (median 2.0 [interquartile range 2.0–4.0] vs 2.0 [1.0–3.0], $p = 0.031$) while rates of IV thrombolysis were similar between groups. Notably, after conducting the propensity score matching, 90-day PSOM and mRS scores did not differ significantly between patients with and without cardiac abnormalities (eTable 1).

Multivariable Predictors of Poor PSOM Outcome at 90 Days

To explore whether association of EVT with outcome differed by cardiac status, stratified models were constructed (Table 2). Among children with cardiac abnormalities, EVT was associated with a significantly lower risk of poor PSOM outcome (OR 0.47, 95% CI 0.26–0.97, $p = 0.018$). In this subgroup, younger age (OR 0.77, 95% CI 0.50–0.91, $p = 0.008$) and higher ASPECTS (OR 0.53, 95% CI 0.24–0.74, $p < 0.001$) were also significantly associated with better outcome.

Among children without cardiac abnormalities, EVT was also independently associated with improved outcome (OR 0.32, 95% CI 0.13–0.74, $p = 0.012$). In this group, among the other variables, only the baseline stroke severity (pedNIHSS) was significantly associated with better outcome (OR 1.14, 95% CI 1.02–1.36, $p = 0.012$).

Interaction Analysis Using Firth Penalized Regression

Given the relatively small sample size and potential for sparse data within subgroups, a Firth penalized logistic regression model was used to formally assess effect modification. After adjustment for age, baseline pedNIHSS score, and ASPECTS, EVT remained independently associated with a reduced risk of poor PSOM outcome (OR 0.24, 95% CI 0.07–0.77, $p = 0.016$). The EVT × cardiac abnormality interaction term was not statistically significant (OR 0.86, 95% CI 0.06–7.99), indicating no evidence that the treatment effect differed on the odds ratio scale by cardiac abnormality status. It should be noted that interaction analyses were prespecified but should be considered primarily exploratory, given the modest sample size and the likelihood of sparse data within some strata.

Ordinal mRS Shift Analysis

Ordinal logistic regression modeling of 90-day mRS scores demonstrated a favorable shift associated with EVT ($\beta = -0.73$, $p = 0.05$). Baseline pedNIHSS score was strongly associated with worse mRS outcomes while higher ASPECTS was independently associated with better outcomes. Cardiac abnormality status and the EVT × cardiac abnormality interaction were not significantly associated with ordinal outcome shifts, with p values of 0.348 and 0.896, respectively, further supporting comparable functional trajectories across groups (Figure 3). However, given limited power for interaction testing, these null findings should likewise be interpreted cautiously.

Longitudinal Neurologic Recovery

Longitudinal mixed-effects modeling of pedNIHSS scores across 3 time points (baseline, 12–24 hours, and discharge) demonstrated significant neurologic improvement over time ($\beta = -3.89$ per time point, $p < 0.001$). EVT significantly modified recovery trajectories, with EVT-treated patients exhibiting steeper improvement compared with non-EVT patients (time × EVT interaction $\beta = -5.33$, $p = 0.001$). These effects remained robust after adjustment for age and ASPECTS.

In an exploratory analysis of change in pedNIHSS scores from admission to follow-up, EVT was associated with a greater absolute reduction in neurologic deficit ($\beta = -4.24$, $p = 0.006$). The EVT × cardiac abnormality term suggested a larger magnitude of improvement in children with cardiac abnormalities, corresponding to an estimated additional reduction in pedNIHSS scores ($p = 0.006$); however, this model demonstrated multicollinearity and was interpreted cautiously as hypothesis-generating rather than confirmatory (Figure 4).

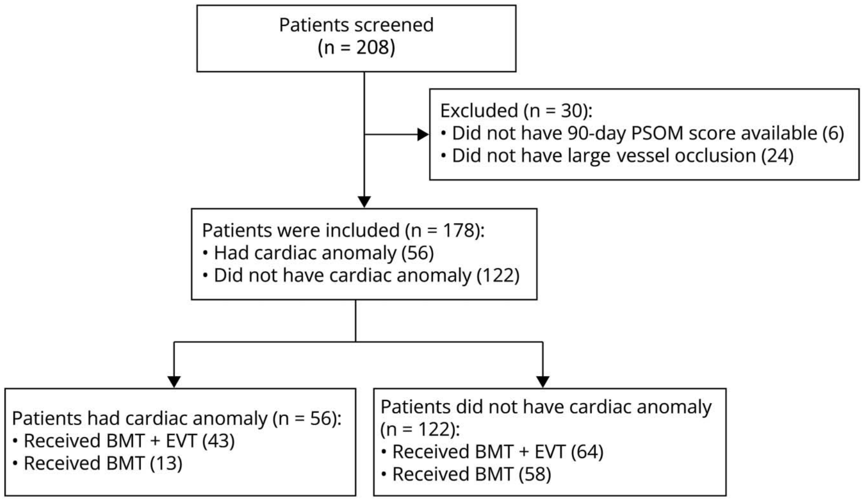
Safety and Adverse Events

Adverse events, such as hemorrhage and need for hemicraniectomy, did not differ significantly according to EVT status in either subgroup (eTable 2). Among children with cardiac abnormalities, adverse events occurred in 7 of 36 patients (19.4%) who underwent EVT compared with 2 of 11 patients (18.2%) managed without EVT, yielding no significant difference (OR 1.07, $p = 1.00$). Similarly, in children without cardiac abnormalities, adverse events were observed in 10 of 54 patients (18.5%) treated with EVT and 8 of 50 patients (16.0%) treated without EVT, with no statistically significant association (OR 1.16, $p = 0.80$). These findings suggest that EVT was not associated with an increased risk of adverse events, regardless of underlying cardiac abnormality status.

Classification of Evidence

This study provides class IV evidence that endovascular therapy in pediatric patients with AIS is associated with

Figure 1 Flowchart of Patient Inclusion



BMT = best medical treatment; EVT = endovascular treatment; PSOM = Pediatric Stroke Outcome Measure.

improved functional outcome irrespective of underlying cardiac abnormality status.

Discussion

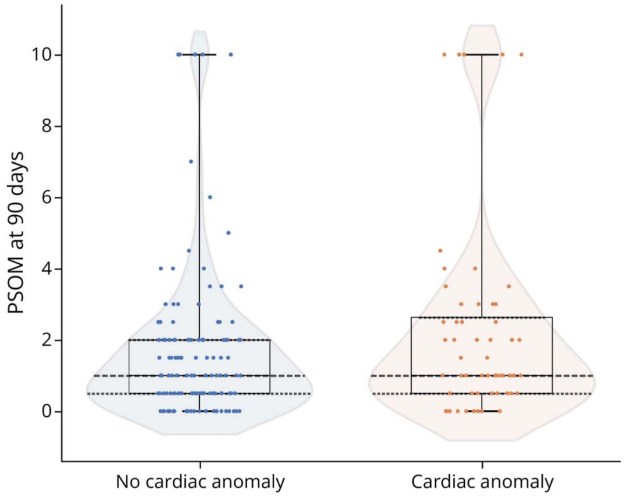
In the previously published primary analysis of the Save ChildS Pro cohort, we observed improved 90-day functional outcomes with EVT vs BMT,[10] but questions persisted regarding the generalizability of those findings to the subgroups of children with cardiac disease. In this study, we examined whether cardiac abnormalities affect the relationship between EVT and functional outcomes in pediatric AIS. In this study, we found that cardiac abnormality status was not independently associated with worse outcomes after adjustment

Table 1 Patient Characteristics

Variable	No cardiac abnormality	Cardiac abnormality	*p* Value
Age	9.00 (6.00–15.00) (n = 121)	7.50 (3.00–12.00) (n = 56)	0.020
Sex (male/total) (%)	70/122 (57.4)	31/56 (55.4)	0.772
Any antithrombotic therapy (%)	7/122 (5.7)	28/56 (50)	<0.001
Anticoagulant (UFH/LMWH/warfarin) (%)	5 (4.1)	19 (33.9)	<0.001
Antiplatelet (aspirin/GP IIb/IIIa) (%)	2 (1.6)	9 (16.1)	<0.001
ASPECTS	8.00 (6.00–9.00) (n = 115)	7.00 (6.00–8.00) (n = 56)	0.084
Time to treatment (minutes)	335.00 (192.00–598.50) (n = 71)	389.50 (271.25–888.25) (n = 38)	0.189
Anterior circulation (vs. posterior)	100/122	47/56	0.914
IV thrombolysis (%)	25/122 (20)	6/56 (10)	0.166
Endovascular treatment (%)	64/122 (52.5)	43/56 (76.8)	0.003
pedNIHSS score at admission	12.00 (8.00–17.00) (n = 122)	12.00 (8.00–18.00) (n = 56)	0.308
General anesthesia (%)	47/62 (75.8)	34/42 (81.0)	0.117
Length of stay (d)	12.00 (7.00–19.00) (n = 115)	15.00 (10.00–28.00) (n = 55)	0.006
PSOM score at 90 d	1.00 (0.50–2.00) (n = 122)	1.00 (0.50–2.62) (n = 56)	0.224
mRS score at 90 d	1.00 (0.00–2.00) (n = 122)	2.00 (1.00–3.00) (n = 56)	0.036

Abbreviations: ASPECTS = Alberta Stroke Program Early CT Score; GP = glycoprotein; LMWH = low molecular weight heparin; mRS = modified Rankin Scale; pedNIHSS = Pediatric National Institutes of Health Stroke Scale; PSOM = Pediatric Stroke Outcome Measure; UFH = unfractionated heparin.
Categorical variables are presented as n (%) and continuous variables as median (interquartile range). Continuous data were compared using the Mann-Whitney *U* test, and categorical data were compared using χ^2 analysis.

Figure 2 Distribution of 90-Day Pediatric Stroke Outcome Measure (PSOM) Scores in Patients With and Without Cardiac Abnormalities



The median is shown by the central line, the interquartile range (interquartile range; 25th–75th percentiles) by the box, and the minimum-maximum by the whiskers.

for baseline clinical and imaging severity. It is important to note that we found no evidence that cardiac abnormalities attenuate the benefit of EVT nor that EVT is associated with excess harm in this high-risk subgroup. These findings are particularly relevant, given that children with cardiac abnormalities represent one of the most clinically challenging populations in pediatric stroke, requiring concurrent anticoagulation and carrying greater procedural and hemodynamic risks.[8,13,14]

Notably, the use of EVT in pediatric AIS has increased substantially, driven by accumulating evidence of technical feasibility, favorable recanalization rates, and acceptable safety profiles in selected children.[15,16] Although randomized controlled trials remain lacking in this population, multicenter registries and observational cohorts have consistently demonstrated outcomes that parallel those observed in adult practice, particularly when treatment is delivered in experienced centers.[17,18] This expanding utilization reflects growing clinical confidence in EVT as a rescue or definitive therapy for large vessel occlusion in children, despite the unique anatomical, physiologic, and etiologic considerations that distinguish pediatric stroke from its adult counterpart. Among these considerations, cardiac abnormalities represent one of the most common and heterogeneous sources of embolic stroke in childhood.[4,8]

Children with cardiac abnormalities represent one of the most clinically challenging subgroups in pediatric stroke. Concerns regarding increased risk of recurrent embolic events, altered clot composition, anticoagulation status, hemodynamic instability, and procedural risk have contributed to therapeutic hesitation in this population.[19] Against this background, our results are clinically reassuring. Across multiple analytic approaches, EVT was consistently associated with a lower risk of poor 90-day functional outcome. In stratified analyses, EVT conferred benefit both in children with cardiac abnormalities and in those without, with comparable directions and magnitudes of effect.

We observed that children with cardiac abnormalities experienced longer hospital stays. Despite this, functional outcomes were similar to those observed in children without cardiac disease. This dissociation between procedural complexity and functional outcome is noteworthy. It suggests that, while EVT in children with cardiac abnormalities may demand greater technical effort or periprocedural resources, these challenges do not translate into worse neurologic recovery. This finding may help counter bias against aggressive reperfusion strategies in this subgroup and supports the feasibility of EVT in experienced centers. It is important to note that although, in our cohort, children with cardiac abnormalities more frequently received antithrombotic/anticoagulation therapy, we did not observe a higher rate of bleeding complications in this subgroup.

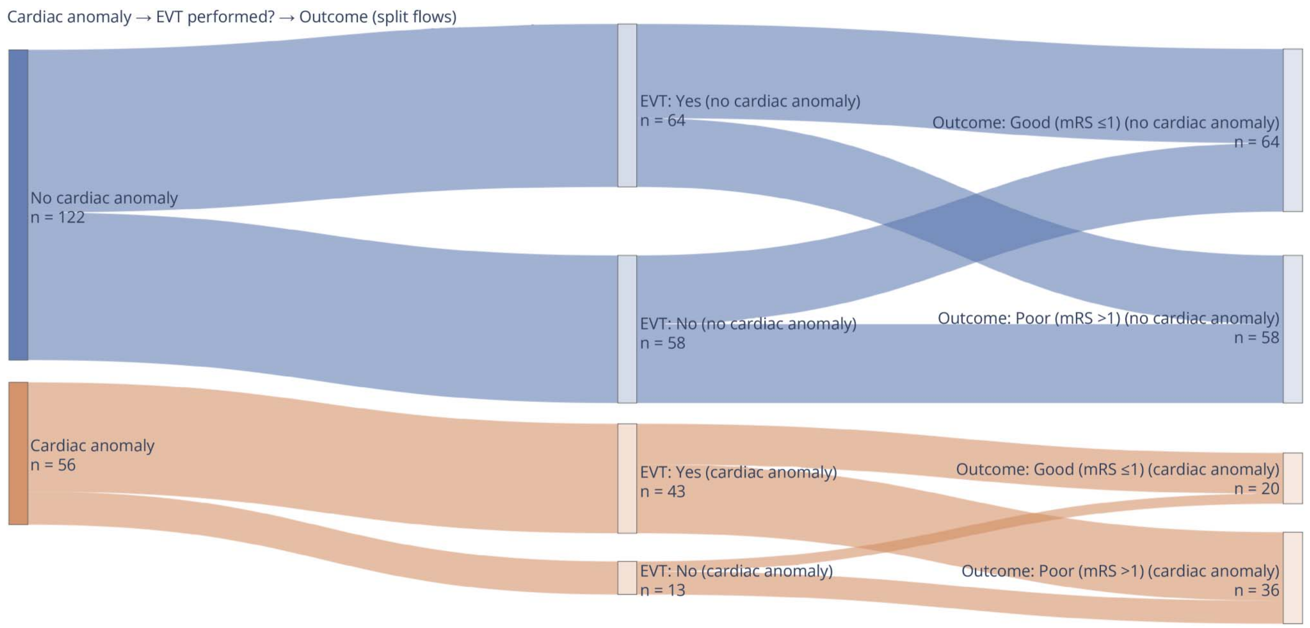
Also worthy of mention, children with cardiac abnormalities represent a heterogeneous subgroup of pediatric stroke

Table 2 Association Between Variables and Poor PSOM Outcomes in Patients With and Without Cardiac Abnormality

Variable	No cardiac abnormality OR (95% CI)	No cardiac abnormality *p* Value[a]	Cardiac abnormality OR (95% CI)	Cardiac abnormality *p* Value[a]
Age	0.93 (0.82–1.02)	0.132	0.77 (0.5–0.91)	0.008
pedNIHSS score at admission	1.14 (1.02–1.36)	0.012	1.07 (0.98–1.32)	0.13
ASPECTS	0.72 (0.33–1.23)	0.242	0.53 (0.24–0.74)	<0.001
Endovascular treatment	0.32 (0.13–0.74)	0.012	0.47 (0.26–0.97)	0.018
IV thrombolysis	0.74 (0.28–2.02)	0.576	0.7 (0.34–1.58)	0.496

Abbreviations: ASPECTS = Alberta Stroke Program Early CT Score; OR = odds ratio; pedNIHSS = Pediatric National Institutes of Health Stroke Scale; PSOM = Pediatric Stroke Outcome Measure.
[a] *p* Value derived from bootstrap.

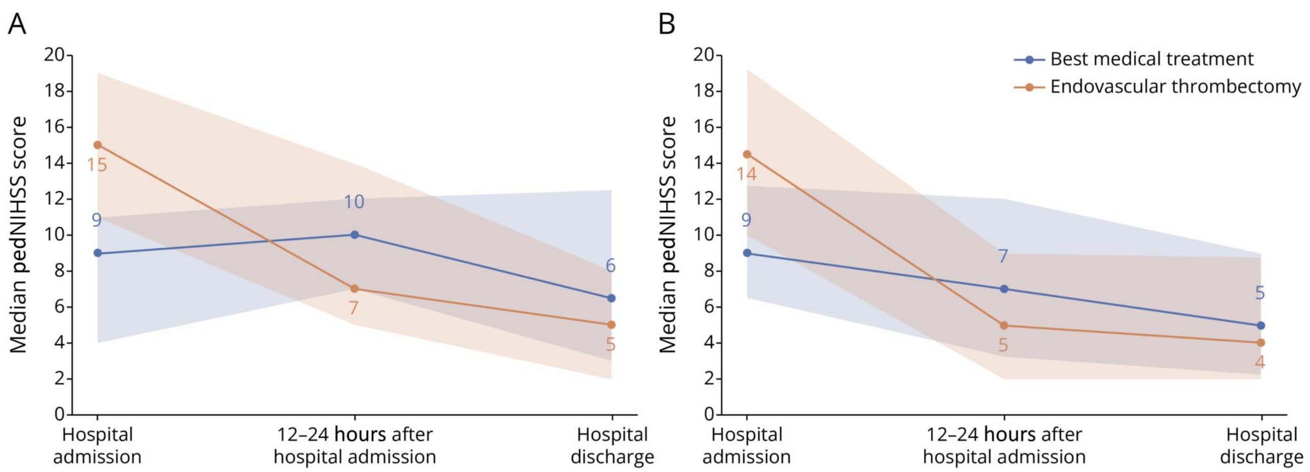
Figure 3 Sankey Diagram Showing the Flow of Patients From Cardiac Abnormality Status (Left) to Whether Endovascular Treatment (EVT) Was Performed (Middle) and to 90-Day Functional Outcome (Right), Dichotomized as Good (mRS 0–1) vs Poor (mRS 2–6)



Band width is proportional to the number of patients, and counts (n) are displayed within each node/flow. mRS = modified Rankin Scale.

patients, and the underlying cardiac pathology may influence both stroke mechanism and acute treatment considerations. Simple abnormalities such as patent foramen ovale or atrial septal defects are often associated with paradoxical embolism and may resemble embolic stroke mechanisms seen in otherwise healthy children, potentially making them favorable candidates for reperfusion therapies.[20,21] By contrast, complex congenital heart diseases, including single-ventricle physiology, cyanotic heart disease, cardiomyopathies, or those requiring previous surgical correction, are frequently associated with chronic anticoagulation, altered hemodynamics, endothelial dysfunction, and higher baseline procedural risk.[22] These factors may raise concerns regarding hemorrhagic complications, vascular access, or anesthesia management, which can influence decision making around EVT and intravenous thrombolysis (IVT).[23,24] Future studies with granular phenotyping of cardiac abnormalities are needed to better delineate which subgroups derive the greatest benefit from EVT and how periprocedural risk can be optimized.

Figure 4 Longitudinal Changes in Pediatric NIH Stroke Scale (pedNIHSS) Scores Among Patients (A) With and (B) Without Cardiac Abnormalities



Finally, the findings of this study further stress the need for the development and implementation of standardized, pediatric-specific protocols for endovascular stroke management.[25,26] Unlike adult stroke care, which is guided by well-established workflows and evidence-based treatment algorithms, pediatric AIS remains characterized by substantial heterogeneity in triage, imaging selection, treatment thresholds, and post-procedural care across centers.[20,27] This variability is particularly pronounced in children with complex comorbidities such as cardiac abnormalities, in whom concerns regarding procedural risk, anticoagulation status, and peri-interventional management may delay or preclude timely reperfusion therapy.[28,29] Furthermore, many of these children may already be sedated, such as in the post–cardiac surgery period while on bypass, which represents a high-risk window for stroke and often leads to delayed recognition of stroke symptoms.[30] The absence of unified protocols can contribute to treatment delays, inconsistent patient selection, and inequitable access to EVT, all of which may adversely affect neurologic outcomes.

Despite the findings, we acknowledge important limitations. As an observational study, residual confounding cannot be excluded, particularly with respect to unmeasured factors such as center-level expertise. Second, although posterior circulation strokes were included in the overall cohort, subgroup analyses comparing anterior vs posterior circulation were not feasible because of limited sample sizes and incomplete reporting across centers. Similarly, medium vessel occlusions were excluded from subgroup analyses because of their low frequency and heterogeneity, which would have compromised the interpretability of treatment effects. As a result, the present findings primarily reflect large vessel anterior circulation strokes. Third, the number of patients receiving IVT was insufficient to allow robust assessment of bridging therapy or interaction effects between IVT and EVT, precluding firm conclusions regarding its additive benefit in this pediatric population. Moreover, sample size constraints limited power for detecting modest interaction effects, and null interaction findings should, therefore, be interpreted as absence of evidence rather than definitive equivalence. The modest sample size also limits power to detect uncommon adverse events; thus, the observed adverse event rates should be interpreted cautiously and ideally confirmed in larger cohorts. In addition, because most patients were treated at high-volume tertiary centers, generalizability to lower volume settings may be limited, and potential EVT selection bias, especially in children with cardiac disease, may have influenced observed associations. Finally, while PSOM is a pediatric-appropriate, domain-based outcome measure, our use of PSOM >0 as a dichotomous end point is intentionally sensitive and may capture minimal deficits that would not be considered "poor" in other frameworks. This choice may increase apparent poor-outcome rates and reduce comparability with studies using stricter cutoffs (e.g., higher PSOM thresholds or mRS-based definitions) and should be considered when interpreting effect estimates.

In this cohort of children with AIS due to large vessel occlusion, EVT was associated with improved functional outcomes and accelerated neurologic recovery, with no evidence that cardiac abnormality status modifies association of treatment with outcome. Despite greater procedural complexity, children with cardiac abnormalities achieved outcomes comparable to those without cardiac disease. These findings support the inclusion of children with cardiac abnormalities in EVT consideration and highlight the need for prospective, multicenter studies to further refine treatment selection in this high-risk population.

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Study Funding

The authors report no targeted funding.

Disclosure

The authors report no relevant disclosures. Go to Neurology.org/N for full disclosures.

Publication History

Received by *Neurology*® January 30, 2026. Accepted in final form May 18, 2026. Submitted and externally peer reviewed. The handling editor was Associate Editor Courtney Wusthoff, MD, MS.

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